第六章补充题参考答案

5.（补充题 1）已知粒子在二维无限深方势阱$V=\begin{cases}0, & 0\le x\le a, 0\le y\le a,\\ \infty, & x<0, x>a, y<0, y>a\end{cases}$中运动,

（1）讨论基态和第一激发态的简并度,并写出相应的能级和能量本征函数;

（2）若加上微扰$H^{'}=\lambda xy$,求最低的两个能级的一级修正.

【解答】

（1）粒子在二维无限深方势阱$V=\begin{cases}0, & 0\le x\le a, 0\le y\le a,\\ \infty, & x<0, x>a, y<0, y>a\end{cases}$中运动，可采用分离变量法得到其能级和能量本征函数:

$$E_{n_1n_2}^{(0)}=\frac{\pi^2\hbar^2}{2ma^2}\left(n_1^2+n_2^2\right),\ n_1,n_2=1,2,3,\cdots$$

$$\psi_{n_1n_2}^{(0)}=\begin{cases}\dfrac{2}{a}\sin\dfrac{n_1\pi x}{a}\sin\dfrac{n_2\pi y}{a}, & 0\le x\le a, 0\le y\le a,\\ 0, & x<0, x>a, y<0, y>a.\end{cases}$$

基态是非简并的，能级为 $E_{11}^{(0)}=\dfrac{\pi^2\hbar^2}{ma^2}$,

本征函数为 $\psi_{11}^{(0)}=\dfrac{2}{a}\sin\dfrac{\pi x}{a}\sin\dfrac{\pi y}{a}$.

第一激发态是二重简并的，能级为$E_{12}^{(0)}=\dfrac{5\pi^2\hbar^2}{2ma^2}$,

本征函数为

$$\psi_{12}^{(0)}=\frac{2}{a}\sin\frac{\pi x}{a}\sin\frac{2\pi y}{a}=\psi_\alpha,$$
$$\psi_{21}^{(0)}=\frac{2}{a}\sin\frac{2\pi x}{a}\sin\frac{\pi y}{a}=\psi_\beta.$$

（2）基态能级的一级修正等于$H^{'}$的平均值，即

$$E_{11}^{(1)}=\left\langle\psi_{11}^{(0)}\middle|H^{'}\middle|\psi_{11}^{(0)}\right\rangle=\frac{4\lambda}{a^2}\int_0^a\int_0^a xy\sin^2\frac{\pi x}{a}\sin^2\frac{\pi y}{a}dxdy=\frac{\lambda}{4}a^2$$

对于第一激发态，采用简并态微扰论求解.

由ψ_α, ψ_β容易算出$H^{'}$的矩阵元

$$H'_{\alpha\alpha}=H'_{\beta\beta}=\frac{\lambda}{4}a^2,$$
$$H'_{\alpha\beta}=H'_{\beta\alpha}=\frac{256}{81\pi^4}\lambda a^2.$$

在$\{\psi_\alpha,\psi_\beta\}$子空间中，$H'$的矩阵元表示为

$$H'=\frac{\lambda}{4}a^2\begin{pmatrix}1 & \frac{1024}{81\pi^4}\\ \frac{1024}{81\pi^4} & 1\end{pmatrix}$$

解久期方程

$$\begin{pmatrix}\frac{\lambda}{4}a^2-E_{12}^{(1)} & \frac{\lambda}{4}a^2\times\frac{1024}{81\pi^4}\\ \frac{\lambda}{4}a^2\times\frac{1024}{81\pi^4} & \frac{\lambda}{4}a^2-E_{12}^{(1)}\end{pmatrix}=0$$

可得能级的一级修正为

$$E_{12}^{(1)}=H'_{\alpha\alpha}\pm H'_{\alpha\beta}=\frac{\lambda}{4}a^2\left(1\pm\frac{1024}{81\pi^4}\right)$$

6.(补充题 2)各向同性三维谐振子的哈密顿算符为

$$H=\frac{1}{2\mu}\left(p_x^2+p_y^2+p_z^2\right)+\frac{1}{2}\mu\omega^2\left(x^2+y^2+z^2\right).$$

加上微扰$W=-\lambda\left(xy+yz+zx\right)$之后,用微扰理论计算基态及第一激发态的一级能量修正.

【参考公式】在占有数表象上,坐标矩阵元为:

$$\langle m|x|n\rangle=\sqrt{\frac{\hbar}{2\mu\omega}}\left[\sqrt{n+1}\delta_{m,n+1}+\sqrt{n}\delta_{m,n-1}\right]$$

解答：无微扰时，三维谐振子的本征解为

$$E_n^0=\left(n+\frac{3}{2}\right)\hbar\omega$$

$$\psi_{n_1n_2n_3}\left(x,y,z\right)=\varphi_{n_1}\left(x\right)\varphi_{n_2}\left(y\right)\varphi_{n_3}\left(z\right),\quad N=n_1+n_2+n_3$$
$$n_1,n_2,n_3,n=0,1,2,3,\cdots$$

对于能级 E_n^0，其简并度为：

$$f_n=\frac{1}{2}(n+1)(n+2)$$

当 $n=0$ 时, 基态非简并, 即 $E_0^0=\frac{3}{2}\hbar\omega$

$$|0\rangle=\psi_0^0(x,y,z)=\varphi_0(x)\varphi_0(y)\varphi_0(z)$$

基态的一级能量修正为

$$\begin{aligned}W_{00}&=\langle 0|W|0\rangle=\int\psi_0^{0*}(x,y,z)W\psi_0^0(x,y,z)dxdydz\\&=\int\varphi_0^*(x)\varphi_0^*(y)\varphi_0^*(z)\left[-\lambda(xy+yz+zx)\right]\varphi_0(x)\varphi_0(y)\varphi_0(z)dxdydz\\&=0\end{aligned}$$

当 $n=1$ 时, 第一激发态存在 3 度简并, 即

$$E_1^0=\frac{5}{2}\hbar\omega$$

$$|1\rangle=\psi_1^0(x,y,z)=\varphi_0(x)\varphi_0(y)\varphi_1(z)$$

$$|2\rangle=\psi_2^0(x,y,z)=\varphi_0(x)\varphi_1(y)\varphi_0(z)$$

$$|3\rangle=\psi_3^0(x,y,z)=\varphi_1(x)\varphi_0(y)\varphi_0(z)$$

利用公式

$$\langle m|x|n\rangle=\sqrt{\frac{\hbar}{2\mu\omega}}\left[\sqrt{n+1}\delta_{m,n+1}+\sqrt{n}\delta_{m,n-1}\right]=\frac{1}{\alpha}\left[\sqrt{\frac{n+1}{2}}\delta_{m,n+1}+\sqrt{\frac{n}{2}}\delta_{m,n-1}\right]$$

可以求出

$$\begin{aligned}W_{11}&=\langle 1|W|1\rangle=\int\psi_1^{0*}(x,y,z)W\psi_1^0(x,y,z)dxdydz\\&=\int\varphi_0^*(x)\varphi_0^*(y)\varphi_1^*(z)\left[-\lambda(xy+yz+zx)\right]\varphi_0(x)\varphi_0(y)\varphi_1(z)dxdydz\\&=0\end{aligned}$$

同理可求

$$W_{22}=W_{33}=0$$

$$
\begin{aligned}
W_{12} &= \langle 1|W|2\rangle = \int \psi_1^{0*}(x,y,z) W \psi_2^0(x,y,z)\, dxdydz \\
&= \int \varphi_0^*(x)\varphi_0^*(y)\varphi_1^*(z)\left[-\lambda(xy+yz+zx)\right]\varphi_0(x)\varphi_1(y)\varphi_0(z)\, dxdydz \\
&= -\lambda \int \varphi_0^*(x)\varphi_0(x)\,dx \int \varphi_0^*(y) y \varphi_1(y)\,dy \int \varphi_1^*(z) z \varphi_0(z) dz \\
&= -\frac{\lambda\hbar}{2\mu\omega} = -\frac{\lambda}{2\alpha^2} \\
&= -A
\end{aligned}
$$

$$W_{12} = W_{21} = W_{23} = W_{32} = W_{13} = W_{31} = -A$$

式中

$$A = \frac{\lambda}{2\alpha^2},\ \alpha^2 = \frac{\mu\omega}{\hbar}$$

在简并子空间中，能量一级修正满足的久期方程为

$$\begin{vmatrix} -E_1^{(1)} & -A & -A \\ -A & -E_1^{(1)} & -A \\ -A & -A & -E_1^{(1)} \end{vmatrix} = 0$$

化成一元三次方程

$$\left(E_1^{(1)}\right)^3 - 3A^2 E_1^{(1)} + 2A^3 = 0$$

将其变形为

$$\left(E_1^{(1)} - A\right)\left[\left(E_1^{(1)}\right)^2 + AE_1^{(1)} - 2A^2\right] = 0$$

解之得

$$E_{11}^{(1)} = \frac{\lambda}{2\alpha^2};\ E_{12}^{(1)} = \frac{\lambda}{2\alpha^2};\quad E_{13}^{(1)} = -\frac{\lambda}{\alpha^2}$$

此即为第一激发态的一级能量修正.